

Michael Plessner

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michaelplessner.github.io

Education

Pennsylvania State University - PhD in Physics 2022 - 2027

John Randall Shuman Troxell Memorial Scholarship in Physics 2025

Edward A. and Rosemary A. Mebus Graduate Fellowship in Physics 2024

Edward M. Frymoyer Honors Scholarship 2024

Pennsylvania State University - MSc in Physics 2020 - 2022

Paul Berg and Daniel J. Larson distinguished graduate fellowship 2020

The University of Hong Kong - Semester Abroad 2016

Northeastern University - BSc in Physics with Honors 2015 - 2019

Merit scholarship for all four years

Dean's scholarship for excellence in coursework

Presidential global scholarship to study in Hong Kong

Presidential global scholarship to perform research in Italy

Research

PhD work - Penn State University - Supervised by Dr Jacob Bourjaily 2022 - 2027

Broadly interested in the mathematical structures of quantum field theories, including...

On-shell methods, scattering amplitudes, and finite representations of observables,

Gauge theory colour tensors, representation theory and monoidal categories,

Covariant formalisms for massive amplitudes and recursion relations ,

Master integrand bases and function spaces at 2-loops and beyond,

Hopf algebras, cluster algebras, and tropical geometry,

Double copy and colour-kinematics duality.

Research internship - Northeastern University - Dr James Halverson 2019 - 2020

Studied solvability of Diophantines in string index calculations under topological transitions.

Research internship - CMS / CERN - Dr Toyoko Orimoto 2018 - 2019

Performed test beam and timing resolution analyses on ECAL hardware for the HL-LHC.

Research internship - Università degli Studi di Milano - Dr Marcel Di Vece 2017 - 2018

Modeled coulombic “planetary” nanoparticle systems formed by a sputtering cluster source.

Research internship - The University of Hong Kong - Dr Yanjun Tu 2016 - 2017

Used ROOT to simulate backgrounds and search for SUSY in dilepton events at CERN.

Conferences, Schools, and Visit

Graduate school

Prospects in Theoretical Physics (PiTP) Summer School IAS, NJ, USA, 2026

Gong show: “QED from $N=4$ ”

Amplitudes Summer School U. of Southampton, UK, 2026

Invited visit to the IAS at Princeton IAS, NJ, USA, 2026

Invited by Nick Early and Nima Arkani-Hamed to discuss representation theory

Symbology@15 Conference MPP, Munich, DE, 2025

Gong show: “Masses, flavours, and colours in amplitudes”

Loop-The-Loop 2 Workshop on Amplitudes (Online), 2025

Michigan Interdisciplinary Meeting on Amplitudes U. of Michigan, MI, USA, 2025

Gong show: “Graph games from representation theory”

TASI Summer School CU Boulder, CO, USA, 2025

Invited visit to UCT Applied Maths department UCT, Cape Town, SA, 2025

Invited talk: “What can representation theory teach us about gauge theory?”

Strings and Geometry Conference ICTP Trieste, IT, 2025

Mathematical Aspects of Scattering Amplitudes Conference GGI, Florence, IT, 2024

Amplitudes Conference IAS, NJ, USA, 2024

Gong show: “A ridiculous formula for the number of $SU(N)$ gluonic colour factors”

Amplitudes Summer School IAS, NJ, USA, 2024

Amplitudes Conference CERN, FR/CH, 2023

Amplitudes Summer School CERN, FR/CH, 2023

Gong show: “Covariant BCFW for all masses”

Amplitudes Conference Charles University, Prague, CZ, 2022

Amplitudes Summer School Charles University, Prague, CZ, 2022

The Amplitude Games Summer School MITP (Online), 2021

Undergraduate

Quantum Gravity Summer School Bard College, NY, USA 2019

April APS Conference Denver, CO, USA 2019

Invited talk: “Test beam studies of precision timing with the upgraded CMS-ECAL”

Publications and Pre-prints

1. Jacob Bourjaily, **MP**, *Covariant BCFW for all Masses*, In Preparation
2. Jacob Bourjaily, **MP**, Philip Velie, *Rational Representations of Lie Algebras*, In Preparation
3. Jacob Bourjaily, **MP**, Philip Velie, *Tree Amplitudes in Pure Gauge Theory with Charged Matter*, 2604.18672
4. Jacob Bourjaily, **MP**, Cristian Vergu, *The Colour Dependence of Amplitudes*, JHEP 08 (2026) **71** – Published 10 August 2026 – 2512.23806
5. Jacob Bourjaily, **MP**, Cristian Vergu, *The Many Colours of Amplitudes*, 2412.21189
6. Jacob Bourjaily, **MP**, et al, *Gauge-Invariant Double-Copies via Recursion Relations*, Phys. Rev. Lett. **131**, 191601 – Published 8 November 2023 – 2307.02542
7. James Halverson, **MP**, Fabian Ruehle, Jiahua Tian, *Kahler Moduli Stabilization and the Propagation of Decidability*, Phys. Rev. D **101**, 046010 – Published 10 February 2020 – 1911.07835
8. Sarita Marom, **MP**, et al, 'Planetary' silver nanoparticles originating from a magnetron sputter plasma, J. Phys. D: Appl. Phys **52**, 095301 – Published 31 December 2019

Teaching at Penn State University

Introductory Mechanics (PHYS 211)	2024, 2026
Introductory Electromagnetism (PHYS 212)	2022, 2023, 2024
Thermal Physics (PHYS 213)	2022
Wave Physics (PHYS 214)	2022, 2025
Algebra-Based Introduction to Mechanics (PHYS 250)	2025
Electromagnetism (PHYS 400)	2021
Subatomic Physics (PHYS 406)	2022, 2025
Quantum Mechanics I (PHYS 410)	2021, 2025
Graduate Quantum Mechanics II (PHYS 562)	2022
Computational Resources and Curriculum Development	2026

Outreach

- 1) Provided tutoring and career advice to minority students at the HBCU (historically black college / university) North Carolina Central University as part of the PREM (Partnership for Research and Education) program

Graduate Electromagnetism 2024, 2025

Undergraduate Electromagnetism and Quantum Mechanics 2023, 2024, 2025

- 2) Tutored Penn State physics graduate students remediating first-year courses 2025

- 3) Research talks at Penn State's "Primordial Universe and Gravity" (PUG) seminar

"Why (Non-Abelian) Gauge Theory?" 2025

"Correcting Coulomb and Newton with On-Shell Methods" 2024

"Quantum Field Theory Without Fear" 2023

- 4) Research talk to the physics and chemistry club at State Area High School

"A brief history of QFT and why the sky is blue" 2022

- 5) Research talk at Penn State's "Mathematical Aspects of Physics" (MAP) seminar

"Why Good Variables Matter: Representations of Poincare" 2022

- 6) Peer mentor to junior Penn State physics PhD students

Philip Velie 2023-2027

Malek Telfah 2026-2027

Rock Climbing

- Sponsored by Mad Rock Climbing from 2014 - present
- Commercial Routesetter, over a decade of experience
- Instagram - @v10issoft
- **Memorable ascents**
 - *Boulders:*
Mooiste Meisie (v13/8B, South Africa),
Donkeystrength (v12/8A+ FA, Nardouw South Africa),
Rainbow Rocket (v11/8A, France),
Freak Brothers (v11/8A, Switzerland),
All Night Diner (v11/8A FA, Nardouw South Africa),
Monstrous Moonshine (v11/8A FA, USA),
973 to the Dangerzone (v6/7A R FA, Hong Kong)
 - *Routes:*
Salathé Wall on **El Capitan** (5.9 C2, 3500' Yosemite),
Snake Dike on **Half Dome** (5.7 R, 2000' Yosemite),
Time Wave Zero on **El Toro** (5.12a, 2300' Mexico),
Flatirons scramble during the TASI 2025 summer school
(**El Camino Royale** 500' - **Royal Arch** 100' - **Dodge Block** 700')
- **New route development during my PhD**
 - **Pennsylvania - (re-)discovering local gems:**
Contributed to several new and old climbing areas around central Pennsylvania. This included bolting 5.12 sport climbs, cleaning central PA's only 60m cliff, and establishing central PA's hardest boulder (**Localized**, v11 FA).
 - **South Africa - Nardouw, Rocklands:**
Worked with landowners over multiple years to cut roads, develop trails, and publish a digital guidebook. Contributed ~100 new climbs. Landowners are Rooibos tea farmers and climbers now help to supplement their income.
 - **West Virginia - The New River Gorge:**
Due to COVID, for the first year of my PhD I lived in my van at a campground. Aside from doing my coursework at picnic tables and chopping fire wood, I established trad climbs up to 5.12, sport to 5.13, and boulders to v9/7C.